

Question	Answer	Marks
2(a)	<u>sum / total</u> momentum (of a system of bodies) is constant or <u>sum / total</u> momentum before = <u>sum / total</u> momentum after	M1
	for an isolated system / no (resultant) <u>external</u> force	A1
2(b)(i)	$240 \times 16 = 480v$ and so $v = 8.0 \text{ m s}^{-1}$ or (initial momentum =) $240 \times 16 (= 3840 \text{ g m s}^{-1})$ and $v = 3840 / 480 = 8.0 \text{ m s}^{-1}$	A1
2(b)(ii)	$(E_K =) \frac{1}{2} mv^2$	C1
	$\Delta E_K = \frac{1}{2} [(0.24 \times 16^2) - (0.48 \times 8.0^2)]$	C1
	= 15 J	A1
2(c)(i)	$F = (0.24 \times 16) / (2.0 \times 10^{-3})$ or $F = (0.48 \times 8) / (2.0 \times 10^{-3})$	C1
	= 1900 N	A1
	direction: to the left	B1
2(c)(ii)	equal (magnitude)	B1
	opposite (direction)	B1

9702/23

Cambridge International AS & A Level – Mark Scheme

October/November 2024

PUBLISHED

Question	Answer	Marks